

The Metrizable $\mathcal{C}(K)$ Case of Problem 7.3 in arXiv:2602.12934

Automatic Math Discovery Smoke Test

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Abstract

Problem 7.3 of De Bernardi–Russo–Sezgek–Somaglia asks whether $\gamma^*(\mathcal{C}(\mathcal{K})) = 1$ for all (metrizable) compact spaces $\mathcal{K}$. We prove the metrizable case. The argument combines the paper’s zero-dimensional and perfect-metrizable cases with the Cantor–Bendixson decomposition and a Dugundji extension operator.

1 Source of the question

The open problem appears in Problem 7.3 of the source paper. The crop below is taken from the original arXiv PDF, page 38.

$p \mapsto \gamma(\ell_1 \oplus_p \ell_1)$, for instance, is it true that it attains its maximum in $p = 2$?

We now pose a problem concerning function spaces, related to our results in Section 4.1 and Section 6.1.

Problem 7.3. *Is it true that $\gamma^*(\mathcal{C}(\mathcal{K})) = 1$ for all (metrizable) compact topological spaces? What are the values of $\gamma(L_p(\mu))$ and $\gamma^*(L_p(\mu))$ for $p \in (2, \infty)$?*

We now mention just one possible sample problem in renorming theory.

Problem 7.4. *Is there a norm $\|\cdot\|$ on ℓ_2 such that $\gamma((\ell_2, \|\cdot\|)) > \sqrt{2}$?*

2 Result

Theorem 1. *Let $\mathcal{K}$ be a compact metrizable space. Then*

$$\gamma^*(\mathcal{C}(\mathcal{K})) = 1.$$

Consequently, the parenthesized metrizable version of Problem 7.3 has an affirmative answer.

Proof. We use the following two facts from the source paper. First, if L is a zero-dimensional compact space, then $\gamma^*(\mathcal{C}(L)) = 1$. Second, if L is a perfect compact metric space, then $\mathcal{C}(L)$ is separable and octahedral, hence the source paper’s octahedral theorem gives $\gamma^*(\mathcal{C}(L)) = 1$.

Let P be the Cantor–Bendixson perfect kernel of $\mathcal{K}$. If $P = \emptyset$, then $\mathcal{K}$ is compact scattered, hence zero-dimensional, and the first source fact gives the conclusion.

Assume now that $P \neq \emptyset$. Then P is a perfect compact metric space, so $\gamma^*(\mathcal{C}(P)) = 1$. Fix $\varepsilon > 0$. Choose numbers $0 < \eta < \alpha < \varepsilon$. Since $\gamma^*(\mathcal{C}(P)) = 1$, there is a subgroup $D_P \subseteq \mathcal{C}(P)$ which is 2-separated and $(1 + \eta)$ -dense. By the Dugundji extension theorem, there is a norm-one linear extension operator

$$E : \mathcal{C}(P) \longrightarrow \mathcal{C}(\mathcal{K}), \quad (Eg)|_P = g.$$

Let

$$H = \{h \in \mathcal{C}(\mathcal{K}) : h|_P = 0 \text{ and } h(\mathcal{K}) \subseteq 2\mathbb{Z}\}.$$

This is a subgroup of $\mathcal{C}(\mathcal{K})$. Define

$$D = E[D_P] + H.$$

It is also a subgroup. We first check separation. If $Ed + h \in D$ is nonzero and $d \neq 0$, then

$$\|Ed + h\|_\infty \geq \|(Ed + h)|_P\|_\infty = \|d\|_\infty \geq 2,$$

because $h|_P = 0$ and D_P is 2-separated. If $d = 0$ and $h \neq 0$, then h takes a nonzero value in $2\mathbb{Z}$, so $\|h\|_\infty \geq 2$. Thus D is 2-separated.

It remains to show that D is $(1 + \varepsilon)$ -dense. Let $f \in \mathcal{C}(\mathcal{K})$. Choose $d \in D_P$ such that

$$\|f|_P - d\|_\infty \leq 1 + \eta,$$

and set $g = f - Ed$. Then $\|g|_P\|_\infty \leq 1 + \eta$. Hence the compact set

$$A = \{x \in \mathcal{K} : |g(x)| \geq 1 + \alpha\}$$

is disjoint from P .

The open subspace $\mathcal{K} \setminus P$ is locally compact, metrizable, and scattered; in particular it is zero-dimensional. Therefore the compact set $A \subseteq \mathcal{K} \setminus P$ has a compact clopen neighbourhood U with $A \subseteq U \subseteq \mathcal{K} \setminus P$. Since U is compact and zero-dimensional, the even-integer approximation used in the source paper applies on U : take a finite clopen partition of U on which g has oscillation $< \eta$, and on each piece choose a nearest even integer to a representative value. This gives $h_U \in \mathcal{C}(U; 2\mathbb{Z})$ with

$$\|g|_U - h_U\|_\infty \leq 1 + \eta.$$

Extend h_U by zero outside U . Since U is clopen in $\mathcal{K}$ and disjoint from P , the extension h belongs to H .

For $x \in U$ we have $|g(x) - h(x)| \leq 1 + \eta$. For $x \notin U$, the definition of A gives $|g(x) - h(x)| = |g(x)| < 1 + \alpha$. Hence

$$\|f - (Ed + h)\|_\infty = \|g - h\|_\infty < 1 + \varepsilon.$$

Thus D is $(1 + \varepsilon)$ -dense and 2-separated. Since $\varepsilon > 0$ was arbitrary, $\gamma^*(\mathcal{C}(\mathcal{K})) \leq 1$. The reverse inequality $\gamma^*(X) \geq 1$ holds for every normed space, so $\gamma^*(\mathcal{C}(\mathcal{K})) = 1$. $\square$

3 Verification notes

No numerical verification is needed. The proof reduces the problem to three standard and cited ingredients:

- the source paper's result for zero-dimensional compact spaces;
- the source paper's separable octahedral result, applied to perfect compact metric spaces;
- the Dugundji extension theorem for closed subsets of compact metric spaces.

The construction of the subgroup $D = E[D_P] + H$ is explicit enough to verify the two required inequalities directly: restriction to P gives separation when the $C(P)$ component is nonzero, and the even-integer subgroup H gives separation on the scattered remainder.

Remark 1. *This packet does not claim to settle the nonmetrizable compact case in Problem 7.3.*